

Construction of 30m long PSC Girder Bridge over Karoba River
Upazila: Pirgongj, District: Rangpur

Design Load = 3750 KN

Girder length = 580 mm

Jack ID: 804021

Pump ID: 04

$$Y = 8.6909x + 31.183$$

$$x = \frac{Y - 31.183}{8.6909}$$

$$x_{20} = 82.71$$

$$x_{40} = 169.00$$

$$x_{60} = 255.30$$

$$x_{80} = 341.60$$

$$x_{95} = 406.32$$

$$x_{98} = 419.27$$

$$x_{100} = 427.90$$

$$x_{102} = 436.53$$

$$x_{105} = 449.47$$

Jack ID: 804023

Pump ID: 03

$$Y = 8.6619x - 28.054$$

$$x = \frac{Y + 28.054}{8.6619}$$

$$x_{20} = 89.82$$

$$x_{40} = 176.41$$

$$x_{60} = 263.00$$

$$x_{80} = 349.58$$

$$x_{95} = 414.52$$

$$x_{98} = 427.51$$

$$x_{100} = 436.17$$

$$x_{102} = 444.83$$

$$x_{105} = 457.82$$

Design Data:

$$\text{Area of strand} = 140 \text{ mm}^2$$

$$\text{Area of cable, } A_1 = 2660 \text{ mm}^2$$

$$\text{Modulus of Elasticity of cable } E_1 = 197000 \text{ MPa}$$

$$\text{Avg Slip} = 6 \text{ mm}$$

Test Data:

$$\text{Area of strand} = 142.50 \text{ mm}^2$$

$$\text{Area of cable, } A_2 = 2707.50 \text{ mm}^2$$

$$\text{Modulus of elasticity, } E_2 = 200400 \text{ MPa}$$

Elongation for grip length:

$$\delta = \frac{PL}{AE}$$

$$= \frac{3750 \times 1000 \times 580}{2660 \times 197000}$$

$$= 4.15 \text{ mm}$$

Correction factor:

$$\frac{A_1 E_1}{A_2 E_2} = \frac{2660 \times 197000}{2707.5 \times 200400}$$

$$= 0.966$$